

# Zhan Ling

Research Scientist @ ByteDance

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## RESEARCH INTEREST

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I am an explorer at heart, dedicated to pushing the frontiers of artificial intelligence. My long-term vision is to develop open-world, self-improving intelligent foundation models that drive us toward super-intelligence and expand the boundaries of human knowledge.

My career is defined by a relentless pursuit of diverse and challenging domains. During my Ph.D., I ventured broadly across imitation learning, reinforcement learning, robotics, 3D representation learning, and LLM reasoning. After that, I continued this exploration by driving LLM agent post-training at the cutting edge. I am continually exploring new technical frontiers, including training long-context LLM agents via reinforcement learning, and developing generative reward models for music generation.

## EDUCATION

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**University of California San Diego**, La Jolla, California, USA 2019.9–2024.12  
Doctor of Philosophy, Computer Science, Advisor: Prof. Hao Su  
Master of Science, Computer Science

**Tsinghua University**, Beijing, China 2015.8–2019.7  
Bachelor of Engineering, Institute for Interdisciplinary Information Sciences (IIIS)  
**Yao Class** (founded by Turing Award Laureate, Prof. Andrew Chi-Chih Yao)

## PUBLICATIONS AND PREPRINTS

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(\*, \*\*, \*\*\* indicate equal contribution)

1. [Seed2.0 Model Card: Towards Intelligence Frontier for Real-World Complexity](#). Bytedance Seed. Preprint.
2. [Natural Language Actor-Critic: Scalable Off-Policy Learning in Language Space](#). Joey Hong, Kang Liu, **Zhan Ling**, Jiecao Chen, Sergey Levine. Preprint.
3. [Scaling Long-Horizon LLM Agent via Context-Folding](#). Weiwei Sun, Miao Lu, **Zhan Ling**, Kang Liu, Xuesong Yao, Yiming Yang, Jiecao Chen. *International Conference on Machine Learning (ICML) 2026*.
4. [Scaling LLM Multi-turn RL with End-to-end Summarization-based Context Management](#). Miao Lu, Weiwei Sun, Weihua Du, **Zhan Ling**, Xuesong Yao, Kang Liu, Jiecao Chen. *Annual Meeting of the Association for Computational Linguistics (ACL) 2026, Main*.
5. [Seed1.8 Model Card: Towards Generalized Real-World Agency](#). Bytedance Seed. Preprint.
6. [Generalizable End-to-End Tool-Use RL with Synthetic CodeGym](#). Weihua Du, Hailei Gong, **Zhan Ling**, Kang Liu, Lingfeng Shen, Xuesong Yao, Yufei Xu, Dingyuan Shi, Yiming Yang, Jiecao Chen. *International Conference on Learning Representations (ICLR) 2026*.

7. [ORIC: Benchmarking Object Recognition in Incongruous Context for Large Vision-Language Models](#). Zhaoyang Li\*, **Zhan Ling\***, Yuchen Zhou, Litian Gong, Erdem Bıyık, Hao Su. *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) 2026*.
8. [MIR-Bench: Benchmarking LLM’s Long-Context Intelligence via Many-Shot In-Context Inductive Reasoning](#). Kai Yan, **Zhan Ling**, Kang Liu, Yifan Yang, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, Jiecao Chen. *Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track 2025*.
9. [LongReason: A Synthetic Long-Context Reasoning Benchmark via Context Expansion](#). **Zhan Ling**, Kang Liu, Kai Yan, Yifan Yang, Weijian Lin, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, Jiecao Chen. Preprint.
10. [ActPlan-1K: Benchmarking the Procedural Planning Ability of Visual Language Models in Household Activities](#). Ying Su, **Zhan Ling**, Haochen Shi, Cheng Jiayang, Yauwai Yim, Yangqiu Song. *Empirical Methods in Natural Language Processing (EMNLP) 2024 Main*.
11. [Unleashing the Creative Mind: Language Model As Hierarchical Policy For Improved Exploration on Challenging Problem Solving](#). **Zhan Ling**, Yunhao Fang, Xuanlin Li, Tongzhou Mu, Mingu Lee, Reza Pourreza, Roland Memisevic, Hao Su. Preprint.
12. [DreamFuser: Value-guided Diffusion Policy for Offline Reinforcement Learning](#). Kairong Luo, Caiwei Xiao, Zhiao Huang, **Zhan Ling**, Yunhao Fang, Hao Su. Preprint.
13. [Deductive Verification of Chain-of-Thought Reasoning](#). **Zhan Ling\***, Yunhao Fang\*, Xuanlin Li, Zhiao Huang, Mingu Lee, Roland Memisevic, Hao Su. *Neural Information Processing Systems (NeurIPS) 2023*.
14. [On the Efficacy of 3D Point Cloud Reinforcement Learning](#). **Zhan Ling\***, Yunchao Yao\*, Xuanlin Li, Hao Su. Preprint.
15. [Distilling Large Vision-Language Model with Out-of-Distribution Generalizability](#). Xuanlin Li\*, Yunhao Fang\*, Minghua Liu, **Zhan Ling**, Zhuowen Tu, Hao Su. *IEEE / CVF International Conference on Computer Vision (ICCV) 2023*.
16. [Reparameterized Policy Learning for Multimodal Trajectory Optimization](#). Zhiao Huang, Litian Liang, **Zhan Ling**, Xuanlin Li, Chuang Gan, Hao Su. *Deep Reinforcement Learning Workshop, Neural Information Processing Systems (NeurIPS) 2022; International Conference on Machine Learning (ICML) 2023, Oral*.
17. [PartSLIP: Low-Shot Part Segmentation for 3D Point Clouds via Pretrained Image-Language Models](#). Minghua Liu, Yinhao Zhu, Hong Cai, Shizhong Han, **Zhan Ling**, Fatih Porikli, Hao Su. *IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR) 2023*.
18. [ManiSkill2: A Unified Benchmark for Generalizable Manipulation Skills](#). Jiayuan Gu\*, Fanbo Xiang\*, Xuanlin Li\*\*, **Zhan Ling\*\***, Xiqiang Liu\*\*, Tongzhou Mu\*\*, Yihe Tang\*\*, Stone Tao\*\*, Xinyue Wei\*\*, Yunchao Yao\*\*, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, Hao Su. *International Conference on Learning Representations (ICLR) 2023*.
19. [Frame Mining: a Free Lunch for Learning Robotic Manipulation from 3D Point Clouds](#). Minghua Liu\*, Xuanlin Li\*, **Zhan Ling\***, Yangyan Li, Hao Su. *Conference on Robot Learning (CoRL) 2022*.
20. [Improving policy optimization with generalist-specialist learning](#). Zhiwei Jia, Xuanlin Li, **Zhan Ling**, Shuang Liu, Yiran Wu, Hao Su. *International Conference on Machine Learning (ICML) 2022*.

21. [Close the Visual Domain Gap by Physics-Grounded Active Stereovision Depth Sensor Simulation](#). Xiaoshuai Zhang\*, Rui Chen\*, Ang Li\*\*, Fanbo Xiang\*\*, Yuzhe Qin\*\*, Jiayuan Gu\*\*, **Zhan Ling\*\***, Minghua Liu\*\*, Peiyu Zeng\*\*, Songfang Han\*\*\*, Zhiao Huang\*\*\*, Tongzhou Mu\*\*\*, Jing Xu, Hao Su. *IEEE Transactions on Robotics (T-RO)* 2023.
22. [Approximate Convex Decomposition for 3D Meshes with Collision-Aware Concavity and Tree Search](#). Xinyue Wei\*, Minghua Liu\*, **Zhan Ling**, Hao Su. *ACM Transactions on Graphics (Proceedings of SIGGRAPH)* 2022.
23. [ManiSkill: Generalizable Manipulation Skill Benchmark with Large-Scale Demonstrations](#). Tongzhou Mu\*, **Zhan Ling\***, Fanbo Xiang\*, Derek Yang\*, Xuanlin Li\*, Stone Tao, Zhiao Huang, Zhiwei Jia, Hao Su. *Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track* 2021.
24. [State Alignment-based Imitation Learning](#). Fangchen Liu, **Zhan Ling**, Tongzhou Mu, Hao Su. *International Conference on Learning Representations (ICLR)* 2020.

## WORK EXPERIENCES

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*ByteDance Inc.*, San Jose

2024.12-Now

Research Scientist

### **End-to-end Multi-turn Function Call Training with Reinforcement Learning**

- Developed a robust synthetic data generation pipeline to significantly enhance end-to-end function call capabilities. ([6])
- Engineered RL pipelines with dynamic context management, successfully empowering LLM agents to execute long-horizon tasks. ([4], [3])

### **Generative Reward Modeling for Music Generation (In Progress)**

- Built an RL pipeline to train a speech-to-text reasoning model as a generative reward model for high-fidelity music generation.

*ByteDance Inc.*, San Jose

2024.6-2024.12

Student Researcher, Host: Jiecao Chen

### **Building a Synthetic Benchmark/Dataset for Long-Context Reasoning**

- Developed a novel algorithm to synthesize high-quality, long-context reasoning question-and-answer pairs.
- Introduced a comprehensive synthetic benchmark to rigorously evaluate and compare the performance of open-source and closed-source models. An internal version of the benchmark was used to evaluate LLMs' long-context reasoning ability. ([9])

*Qualcomm AI Research*, San Diego

2023.6-2023.9

System 2 Team

Interim Engineering Intern, Host: Mingu Lee, Reza Pourreza, Roland Memisevic

### **Unleashing the Creative Mind: Language Model As Hierarchical Policy For Improved Exploration on Challenging Problem Solving**

- Addressed the limited exploration capabilities of LLMs in reasoning tasks by identifying their bottleneck in high-level tactic spaces.
- Developed a novel hierarchical policy framework to enhance exploration: a high-level policy dictates strategic direction while a low-level policy executes detailed reasoning steps. ([11])

*Qualcomm AI Research*, Remote

2023.3-2023.6

System 2 Team

Support Engineering, Host: Mingu Lee, Roland Memisevic

### **Deductive Verification of Chain-of-Thought Reasoning**

- Proposed "Natural Program," a verifiable reasoning format to address LLM hallucinations and logical errors.
- Leveraged LLMs for step-by-step self-verification, successfully filtering invalid chains and boosting reasoning reliability. ([13])

*Qualcomm AI Research*, Remote

2022.6-2023.3

System 2 Team

Interim Engineering Intern, Host: Mingu Lee, Roland Memisevic

### **Learning Code Execution with Language Model**

- Investigated the capability of LLMs to execute code as a Turing machine by training models (up to 6.7B) on Python code execution traces.

- Developed a random code generator supporting loops, conditionals, and function calls to synthesize robust training data.
- Engineered a modified interpreter to automatically extract execution traces from generated code for model training.
- Trained and evaluated LLMs on code execution tasks, achieving high accuracy and strong generalization to unseen programs.
- Resulted in a filed US patent [US20250355632A1].

*X, the moonshot factory*, Remote

2020.6-2020.9

Mineral Project

AI Resident, Host: Lianghao Li, Kangkang Wang

### **Lane Detection for Agricultural Imagery**

- Developed a lane detection algorithm based on the Hough transform in Python, accelerated with Cython.

### **Improving Weed Detection with Lane Detection**

- Built a weed detection model based on EfficientDet.
- Integrated the lane detection algorithm into the weed detection model to improve overall detection performance.

*University of California San Diego*, San Diego

2019.9-2024.12

Su Lab

Graduate Research Assistant, Advisor: Hao Su

### **Embodied AI**

- Co-developed the ManiSkill benchmarks for evaluating the generalization ability of robotic manipulation skills. ([23], [18])
- Developed reinforcement learning and imitation learning algorithms for robotic manipulation tasks. ([24], [20])

### **Computer Vision**

- Developed 3D perception algorithms, including convex decomposition of 3D meshes and 3D part segmentation. ([22], [17])
- Integrated 3D representation learning with reinforcement learning for robotic manipulation. ([19], [14])

### **Reasoning**

- Developed algorithms to improve the reasoning reliability and exploration ability of LLMs. ([13], [11])

## **AWARDS AND SERVICES**

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### **Awards:**

**First Prize (Gold Medal)** in National Olympiad in Informatics (NOI), China, 2014

Outstanding Freshman Scholarship, Tsinghua University, 2015

**Conference Reviewer:** CVPR 2022-2026; ICCV 2021, 2023, 2025; ECCV 2022, 2024, 2026; NeurIPS 2023-2025; ICLR 2024-2026; ICML 2024-2026; ACCV 2024; ICRA 2025; AISTATS 2025; ACL RR 2026.

**Journal Reviewer:** T-RO; RA-L.

**Program Committee:** AAAI 2025-2026.

**Workshop Reviewer:**

Generalizable Policy Learning in the Physical World, ICLR 2022.

Interdisciplinary Exploration of Generalizable Manipulation Policy Learning: Paradigms and Debates, RSS 2023.

**Challenge Organizer:**

[SAPIEN ManiSkill Challenge 2021](#)

[SAPIEN ManiSkill Challenge 2022](#)

## TECHNICAL SKILLS

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Programming Languages: Python, C, C++.

Deep Learning: PyTorch, TensorFlow, JAX, PyTorch CUDA extension.

Large Models: Pre-training data processing, Distributed training, Fine-tuning, Reinforcement learning, Quantization.

Parallel Computing: MPI, OpenMP, CUDA, AVX.

Simulation: SAPIEN, MuJoCo.

Additional Skills: pybind11, ROS.

last updated: July 29, 2026